C33 Chromatography – Knowledge Organiser

AQA A-level Chemistry • Set 1: TLC and Principles **Paper 2**

#	Question	Answer
1	Stationary phase	Phase that does not move
2	Mobile phase	Phase that moves
3	What separation depends on	Balance between solubility in mobile phase and retention by stationary phase
4	TLC stationary phase	Silica or alumina on a plate
5	R_f equation	Distance moved by spot ÷ distance moved by solvent
6	Why the baseline is drawn in pencil	Ink would separate
7	Why the baseline is above the solvent	Spots would dissolve into the solvent
8	Why a lid is used	Saturates the atmosphere with solvent
9	Two ways to locate colourless spots	UV light, iodine (or ninhydrin)
10	How a substance is identified	Compare R_f with known samples

C33 Chromatography – Knowledge Organiser

AQA A-level Chemistry • Set 2: Column Chromatography and GC **Paper 2**

#	Question	Answer
1	Column chromatography stationary phase	Silica or alumina powder
2	GC mobile phase	Unreactive carrier gas (e.g. He, N_2)
3	GC stationary phase	High-boiling liquid on a solid support
4	Retention time	Time from injection to detection
5	Peak area in GC shows	Relative amount of component
6	GC–MS	GC linked to a mass spectrometer
7	Use of GC–MS	Identifies each component
8	Uses of GC	Drug testing, pollution, forensics
9	Why R_f must be compared under the same conditions	Depends on solvent and temperature
10	Why fluorescent TLC plates are used	Spots show dark under UV